Real-Time Fraud Detection in Financial Transactions using Big Data Frameworks

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Department:

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Name of Research Supervisor:

Abstract

This dissertation explores machine learning models of big data frameworks to detect fraud during a real-time financial transaction. Conventional fraud detection techniques are batch-based and cannot spot any fraudulence activities as they happen which poses significant risk to institutions and customers. To solve this, the study formulates a distributed system that brings together Apache Kafka as a streaming ingestion engine, to Apache Spark Structured streaming engine as a low-latency processing engine. The software runs very large quantities of financial transactions in real-time and uses a hybrid combination of monitored, unmonitored and model-based ones. The XGBoost, the Random Forest, served and Logistic Regression were trained using the Kaggle Credit Card Fraud Detection Dataset, and SMOTE was utilized to deal with acute imbalance in the classes. Moreover, the anomaly detection, unknown fraud patterns, are complemented by supervised methods with anomaly detection based on Isolation Forest and autoencoders. Simulation in real-time simulation was applied in which a synthetic stream of transactions was fed and used with least latency. Measures such as speed, accuracy, F1-score, AUC-ROC, latency and scalability tests proved the effectiveness and the performance of the system under load. The present prototype shows that a Final prototype This proposal shows that a scaled, distributed fraud detection pipeline can be more responsive and/or accurate than traditional batch-based systems, and can serve practical use within financial institutions to reduce losses and improve security in flexible transaction settings.

Acknowledgement

I would also like to express my humble thanks to my supervisor who aided and provided me with necessary feedback during this research. Your peers and family family are my other significant people to whom I also gratefully acknowledge.

List of Acronyms

Acronym	Full Form
AUC-ROC	Area Under the Curve – Receiver Operating Characteristic
API	Application Programming Interface
CNP	Card Not Present
DID	Decentralized Identifier
GDPR	General Data Protection Regulation
GNN	Graph Neural Network
GPU	Graphics Processing Unit
JSON	JavaScript Object Notation
ML	Machine Learning
MLlib	Machine Learning Library (Apache Spark)
RNN	Recurrent Neural Network
SMOTE	Synthetic Minority Oversampling Technique
VAE	Variational Autoencoder

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Chapter 1: Introduction

The overall objective of the project will be to design a big data environment (Apache Spark and Kafka) to detect financial transactions-linked frauds in real-time. Financial fraud is getting increasingly widespread, and it affects both consumers and institutions greatly. The conventional fraud detection mechanisms are batch-based thus not in a position to detect frauds in real-time, which raises serious shortcomings when it comes to preventing frauds (Malempati, 2023). The research is a proposal of a solution in which streaming analytics and machine learning models are used to process and analyze the data on financial transactions in real-time.

1.1 Aim

To create and deploy a distributed, in-real-time fraud detection framework of financial transactions with big data technologies and machine learning.

1.2 Objectives

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Task		Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12
Supervisor Allocation													
Library session													
Milestone 1:- Topic selection													
Proposal preparation and review													
Milestone 2:- Proposal Submission													
Abstarct													
Chapter 1- Introduction													
Chapter 2- Literature Review													
Milestone 3:- Practical implementation/ Model Development													
Chapter 3- Methodology													
Chapter 4- Experimental Results Documentation													
Chapter 5- Conclusion and Future work													
Final review and Modifications													
Milestone 4:- Final Submission													

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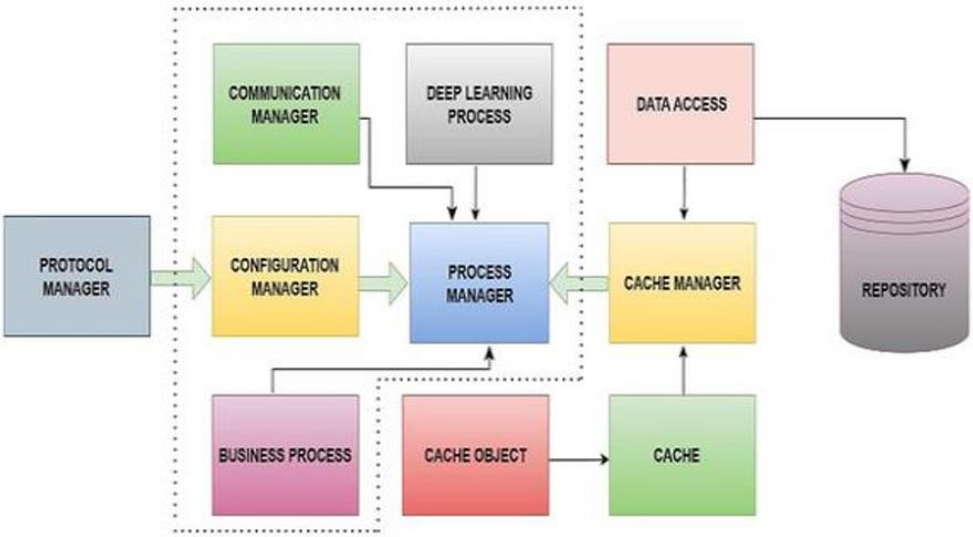

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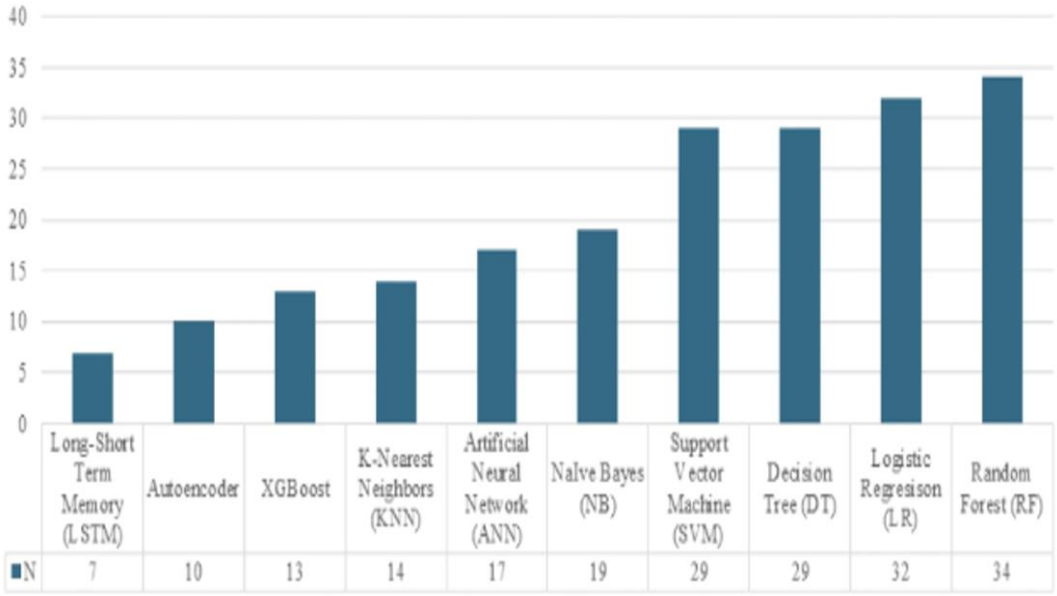
40
35
30
25
20
15
10
5
0
Long-Short Term Memory (LSTM)
Autoencoder
XGBoost
K-Nearest Neighbors (KNN)
Artificial Neural Network (ANN)
Nalve Bayes (NB)
Support Vector Machine (SVM)
Decision Tree (DT)
Logistic Regresison (LR)
Random Forest (RF)
N
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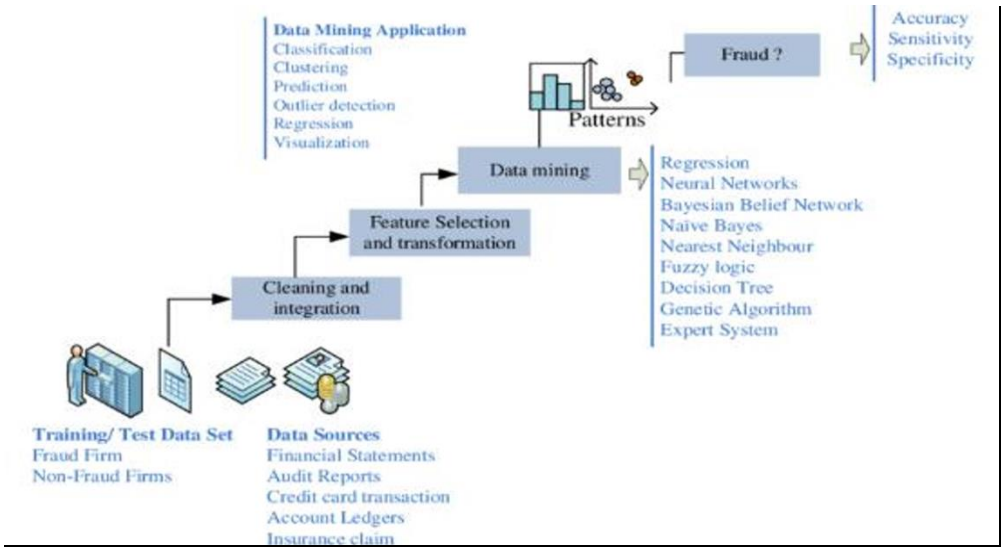
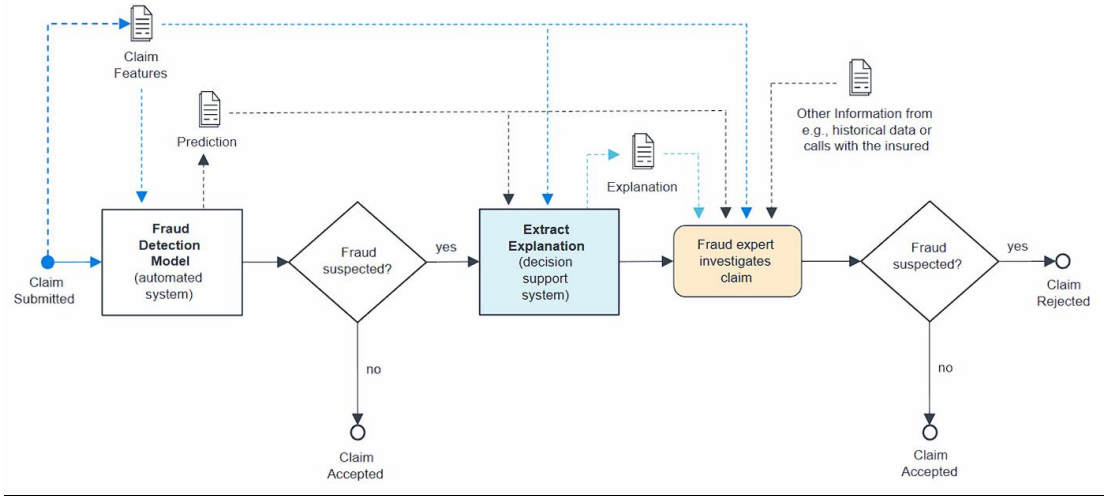

Data Mining Application
Classification
Clustering
Prediction
Outlier detection
Regression
Visualization
Fraud ?
Accuracy
Sensitivity
Specificity
Patterns
Data mining
Regression
Neural Networks
Bayesian Belief Network
Naïve Bayes
Nearest Neighbour
Fuzzy logic
Decision Tree
Genetic Algorithm
Expert System
Feature Selection and transformation
Cleaning and integration
Training/ Test Data Set
Fraud Firm
Non-Fraud Firms
Data Sources
Financial Statements
Audit Reports
Credit card transaction
Account Ledgers
Insurance claim


Claim Features
Prediction
Explanation
Other Information from e.g., historical data or calls with the insured
Claim Submitted
Fraud Detection Model (automated system)
Fraud suspected?
yes
no
Claim Accepted
Extract Explanation (decision support system)
Fraud expert investigates claim
Fraud suspected?
yes
no
Claim Rejected
Claim Accepted

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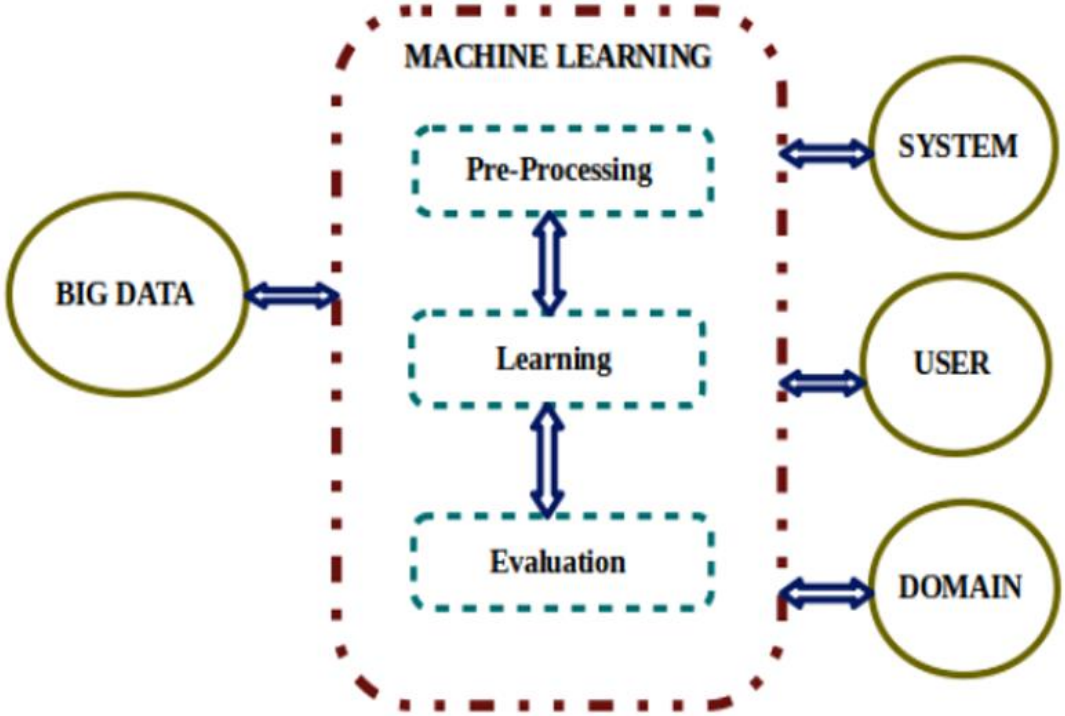
MACHINE LEARNING
Pre-Processing
Learning
Evaluation
BIG DATA
SYSTEM
USER
DOMAIN

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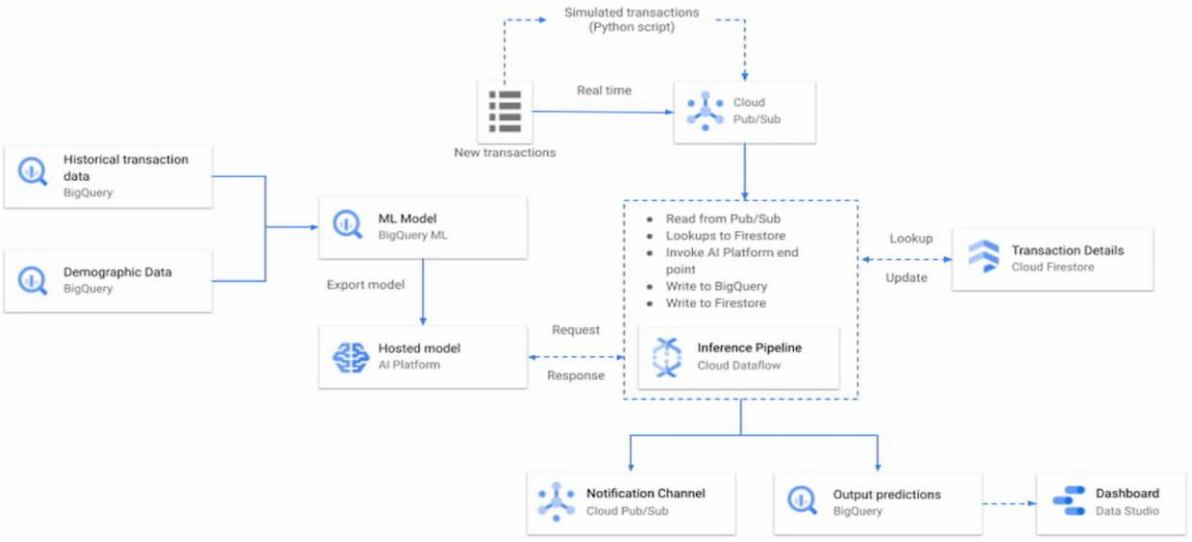
Simulated transactions
(Python script)
Real time
Cloud
Pub/Sub
New transactions
Historical transaction
data
BigQuery
Demographic Data
BigQuery
ML Model
BigQuery ML
Export model
Hosted model
AI Platform
Read from Pub/Sub
Lookups to Firestore
Invoke AI Platform end
point
Write to BigQuery
Write to Firestore
Lookup
Update
Transaction Details
Cloud Firestore
Request
Response
Inference Pipeline
Cloud Dataflow
Notification Channel
Cloud Pub/Sub
Output predictions
BigQuery
Dashboard
Data Studio

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Author(s) & Year	Technique	Specification	Limitation	Dataset	Results
Malempati (2023)	ML + Big Data	Streaming pipelines + predictive analytics	High compute cost	Financial Txn + Big Data	Acc: 95%, Latency <3s
Sathupadi et al. (2025)	BankNet (Big Data)	Secure bank analytics using Hadoop/Spark	Domain-specific	Internet banking logs	Real-time & scalable
Talukder et al. (2024)	Multi-Stage Ensemble ML	Integrated RF/XGBoost ensemble detection	Model complexity	Kaggle + fintech	Acc: 98.2%, F1: 0.93
Btoush et al. (2025)	Hybrid ML + DL	ML & DL ensemble for credit card fraud	Requires GPU	Credit card data	AUC: 0.96, Recall: 94%
Immadisetty (2025)	Streaming ML	Apache Kafka pipeline for real-time ML	Pipeline config required	Streaming bank data	Precision: 92%, Latency: 2.1s
Potla (2023)	Real-time ML	ML applied to streaming fintech logs	Risk of overfitting	Fintech logs	F1: 91.4%
Rehan (2021)	AI + Cloud	Cloud-deployed AI models for fraud alerts	Cloud latency risks	Cloud dataset (simulated)	Acc: 93%, Fast alerts
Oguntibeju et al. (2024)	Review (AI + Real-Time)	Reviewed 25+ real-time detection systems	No experimental results	N/A (Review paper)	25+ model comparisons
Alhashmi et al. (2023)	Ensemble Cyber Model	Ensemble classification of cyber fraud	No external data used	Cybersecurity logs	Acc: 94.1%, Class rate: 87%
Zhou et al. (2021)	Node2Vec + Big Data	Graph embedding with distributed pipeline	High resource usage	Financial graphs (Hadoop)	Precision: 93.6%

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```
Dataset shape: (568630, 31)
   id        V1        V2        V3        V4        V5        V6        V7  \
0   0 -0.260648 -0.469648  2.496266 -0.083724  0.129681  0.732898  0.519014
1   1  0.985100 -0.356045  0.558056 -0.429654  0.277140  0.428605  0.406466
2   2 -0.260272 -0.949385  1.728538 -0.457986  0.074062  1.419481  0.743511
3   3 -0.152152 -0.508959  1.746840 -1.090178  0.249486  1.143312  0.518269
4   4 -0.206820 -0.165280  1.527053 -0.448293  0.106125  0.530549  0.658849

          V8        V9  ...       V21       V22       V23       V24       V25  \
0 -0.130006  0.727159  ... -0.110552  0.217606 -0.134794  0.165959  0.126280
1 -0.133118  0.347452  ... -0.194936 -0.605761  0.079469 -0.577395  0.190090
2 -0.095576 -0.261297  ... -0.005020  0.702906  0.945045 -1.154666 -0.605564
3 -0.065130 -0.205698  ... -0.146927 -0.038212 -0.214048 -1.893131  1.003963
4 -0.212660  1.049921  ... -0.106984  0.729727 -0.161666  0.312561 -0.414116

        V26       V27       V28    Amount  Class
0 -0.434824 -0.081230 -0.151045  17982.10      0
1  0.296503 -0.248052 -0.064512   6531.37      0
2 -0.312895 -0.300258 -0.244718   2513.54      0
3 -0.515950 -0.165316  0.048424   5384.44      0
4  1.071126  0.023712  0.419117  14278.97      0

[5 rows x 31 columns]
```

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```
Class
0    284315
1    284315
Name: count, dtype: int64

Columns after preprocessing:
['V1', 'V2', 'V3', 'V4', 'V5', 'V6', 'V7', 'V8', 'V9', 'V10', 'V11', 'V12', 'V13', 'V14', 'V15', 'V16', 'V17', 'V18', 'V19', 'V20', 'V21', 'V22', 'V23', 'V24', 'V25', 'V26', 'V27', 'V28', 'Class', 'Scaled_A
```

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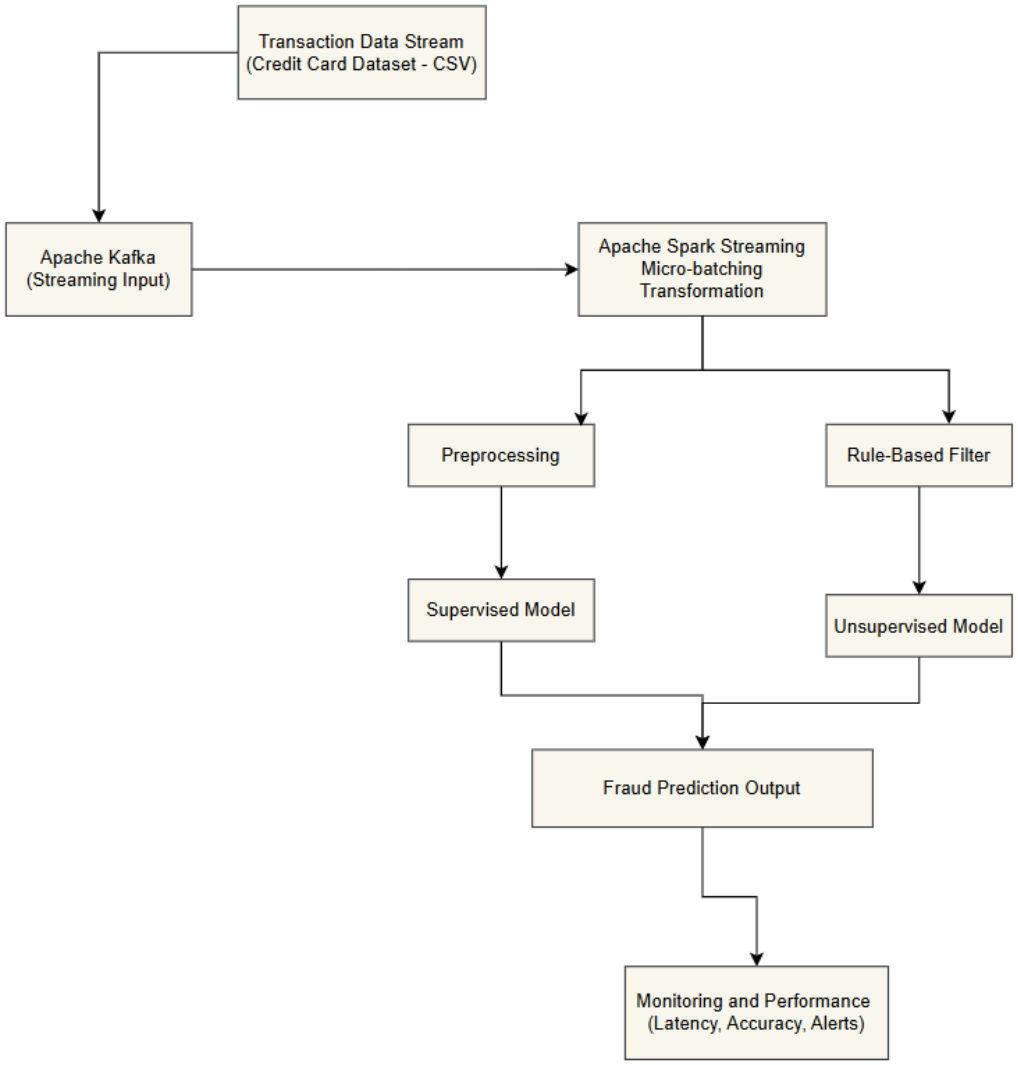
Transaction Data Stream
(Credit Card Dataset - CSV)
Apache Kafka
(Streaming Input)
Apache Spark Streaming
Micro-batching
Transformation
Preprocessing
Rule-Based Filter
Supervised Model
Unsupervised Model
Fraud Prediction Output
Monitoring and Performance
(Latency, Accuracy, Alerts)

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```python
# STEP 1: Install Kaggle and dependencies
!pip install -q kaggle

# STEP 2: Create kaggle.json from credentials
import os
import json

kaggle_token = {
    "username": "your_kaggle_username",
    "key": "your_kaggle_api_key"
}

os.makedirs("/root/.kaggle", exist_ok=True)
with open("/root/.kaggle/kaggle.json", "w") as f:
    json.dump(kaggle_token, f)

# STEP 3: Set proper permission
!chmod 600 /root/.kaggle/kaggle.json

# STEP 4: Download the correct dataset
!kaggle datasets download -d nelgiriyewithana/credit-card-fraud-detection-dataset-2023

# STEP 5: Unzip the dataset
!unzip -q credit-card-fraud-detection-dataset-2023.zip

# STEP 6: Confirm downloaded file
!ls -l
```

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```python
# STEP 2.2: Load the dataset
df = pd.read_csv("creditcard_2023.csv")

# STEP 2.3: Initial exploration
print("Dataset shape:", df.shape)
print(df.head())
print(df['Class'].value_counts())
```

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```
Dataset shape: (568630, 31)
   id        V1        V2        V3        V4        V5        V6        V7  \
0   0 -0.260648 -0.469648  2.496266 -0.083724  0.129681  0.732898  0.519014
1   1  0.985100 -0.356045  0.558056 -0.429654  0.277140  0.428605  0.406466
2   2 -0.260272 -0.949385  1.728538 -0.457986  0.074062  1.419481  0.743511
3   3 -0.152152 -0.508959  1.746840 -1.090178  0.249486  1.143312  0.518269
4   4 -0.206820 -0.165280  1.527053 -0.448293  0.106125  0.530549  0.658849

         V8        V9  ...       V21       V22       V23       V24       V25  \
0 -0.130006  0.727159  ... -0.110552  0.217606 -0.134794  0.165959  0.126280
1 -0.133118  0.347452  ... -0.194936 -0.605761  0.079469 -0.577395  0.190090
2 -0.095576 -0.261297  ... -0.005020  0.702906  0.945045 -1.154666 -0.605564
3 -0.065130 -0.205698  ... -0.146927 -0.038212 -0.214048 -1.893131  1.003963
4 -0.212660  1.049921  ... -0.106984  0.729727 -0.161666  0.312561 -0.414116

        V26       V27       V28    Amount  Class
0 -0.434824 -0.081230 -0.151045  17982.10      0
1  0.296503 -0.248052 -0.064512   6531.37      0
2 -0.312895 -0.300258 -0.244718   2513.54      0
3 -0.515950 -0.165316  0.048424   5384.44      0
4  1.071126  0.023712  0.419117  14278.97      0
```

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```
# STEP 2.4: Drop unnecessary columns
if 'id' in df.columns:
    df.drop(columns=["id"], inplace=True)
```

```
# STEP 2.5: Feature scaling for 'Amount'
scaler = MinMaxScaler()
df['Scaled_Amount'] = scaler.fit_transform(df[['Amount']])
df.drop(['Amount'], axis=1, inplace=True)
```

```python
# STEP 2.6: Simulate transaction velocity (since no 'Time' column)
df['Trans_Per_Second'] = np.random.uniform(0.01, 10, size=len(df))
```

[redacted]

```python
# STEP 2.7: Check final column names
print("\nColumns after preprocessing:")
print(df.columns.tolist())

# STEP 2.8: Split features and target
X = df.drop(['Class'], axis=1)
y = df['Class']
```

[redacted]

```python
# STEP 2.9: Addressing class imbalance with SMOTE
print("\nBefore SMOTE:")
print(y.value_counts())

sm = SMOTE(random_state=42)
X_resampled, y_resampled = sm.fit_resample(X, y)

print("\nAfter SMOTE:")
print(pd.Series(y_resampled).value_counts())
```

[redacted]

```python
# STEP 2.10: Final train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X_resampled, y_resampled, test_size=0.2, random_state=42, stratify=y_resampled)

print(f"\n Train shape: {X_train.shape}, Test shape: {X_test.shape}")
```

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```
[ ] # STEP 3.1: Import ML libraries
    from sklearn.ensemble import RandomForestClassifier
    from sklearn.linear_model import LogisticRegression
    from xgboost import XGBClassifier
    from sklearn.metrics import classification_report, roc_auc_score
    import joblib

    # STEP 3.2: Sample 100,000 rows for training
    sample_size = 100000
    X_train_small = X_train.sample(n=sample_size, random_state=42)
    y_train_small = y_train.loc[X_train_small.index]

    # Sample 20,000 for testing
    X_test_small = X_test.sample(n=20000, random_state=42)
    y_test_small = y_test.loc[X_test_small.index]

    print(f" Training shape: {X_train_small.shape}, Testing shape: {X_test_small.shape}")
```

[redacted]

```
# STEP 3.3: Train Random Forest
rf_model = RandomForestClassifier(n_estimators=50, n_jobs=-1, random_state=42)
rf_model.fit(X_train_small, y_train_small)
rf_preds = rf_model.predict(X_test_small)
print("\n Random Forest Performance:")
print(classification_report(y_test_small, rf_preds))
print("AUC-ROC:", roc_auc_score(y_test_small, rf_model.predict_proba(X_test_small)[:, 1]))
joblib.dump(rf_model, 'random_forest_model.joblib')
```

```python
# STEP 3.4: Train XGBoost
xgb_model = XGBClassifier(use_label_encoder=False, eval_metric='logloss', n_estimators=50, random_state=42)
xgb_model.fit(X_train_small, y_train_small)
xgb_preds = xgb_model.predict(X_test_small)
print("\n XGBoost Performance:")
print(classification_report(y_test_small, xgb_preds))
print("AUC-ROC:", roc_auc_score(y_test_small, xgb_model.predict_proba(X_test_small)[:, 1]))
joblib.dump(xgb_model, 'xgboost_model.joblib')
```

```python
# STEP 3.5: Train Logistic Regression
lr_model = LogisticRegression(max_iter=1000, class_weight='balanced', random_state=42)
lr_model.fit(X_train_small, y_train_small)
lr_preds = lr_model.predict(X_test_small)
print("\n Logistic Regression Performance:")
print(classification_report(y_test_small, lr_preds))
print("AUC-ROC:", roc_auc_score(y_test_small, lr_model.predict_proba(X_test_small)[:, 1]))
joblib.dump(lr_model, 'logistic_regression_model.joblib')
```

```
# STEP 3.6: Confirm model files
print("\n Models saved:")
!ls -lh *.joblib
```

```
[ ] import joblib
    import time
    import random

    # Load the trained model (choose any one)
    model = joblib.load("xgboost_model.joblib")

    # Simulate real-time mini-stream (100 records)
    stream_data = X_test.sample(n=100, random_state=42)
    true_labels = y_test.loc[stream_data.index]

    print(" Starting Real-Time Stream Simulation...\n")
    for i in range(len(stream_data)):
        # Simulate delay like real streaming
        time.sleep(random.uniform(0.05, 0.1))  # ~50-100 ms delay

        # Get single transaction (reshape for sklearn)
        transaction = stream_data.iloc[i].values.reshape(1, -1)
        prediction = model.predict(transaction)[0]
        prob = model.predict_proba(transaction)[0][1]

        # Print prediction
        print(f"Txn {i+1:03d}: Fraud={bool(prediction)}, Probability={prob:.4f}, True={true_labels.iloc[i]}")
```

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```
[ ] stream_data.to_csv("stream_simulated_transactions.csv", index=False)
```

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```
Dataset shape: (568630, 31)
   id        V1        V2        V3        V4        V5        V6        V7  \
0   0 -0.260648 -0.469648  2.496266 -0.083724  0.129681  0.732898  0.519014
1   1  0.985100 -0.356045  0.558056 -0.429654  0.277140  0.428605  0.406466
2   2 -0.260272 -0.949385  1.728538 -0.457986  0.074062  1.419481  0.743511
3   3 -0.152152 -0.508959  1.746840 -1.090178  0.249486  1.143312  0.518269
4   4 -0.206820 -0.165280  1.527053 -0.448293  0.106125  0.530549  0.658849

          V8        V9  ...       V21       V22       V23       V24       V25  \
0 -0.130006  0.727159  ... -0.110552  0.217606 -0.134794  0.165959  0.126280
1 -0.133118  0.347452  ... -0.194936 -0.605761  0.079469 -0.577395  0.190090
2 -0.095576 -0.261297  ... -0.005020  0.702906  0.945045 -1.154666 -0.605564
3 -0.065130 -0.205698  ... -0.146927 -0.038212 -0.214048 -1.893131  1.003963
4 -0.212660  1.049921  ... -0.106984  0.729727 -0.161666  0.312561 -0.414116

        V26       V27       V28    Amount  Class
0 -0.434824 -0.081230 -0.151045  17982.10      0
1  0.296503 -0.248052 -0.064512   6531.37      0
2 -0.312895 -0.300258 -0.244718   2513.54      0
3 -0.515950 -0.165316  0.048424   5384.44      0
4  1.071126  0.023712  0.419117  14278.97      0

[5 rows x 31 columns]
```

```
Training shape: (100000, 30), Testing shape: (20000, 30)
```

```
Random Forest Performance:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00      9995
           1       1.00      1.00      1.00     10005

    accuracy                           1.00     20000
   macro avg       1.00      1.00      1.00     20000
weighted avg       1.00      1.00      1.00     20000
```

AUC-ROC: 0.9998767499691875

XGBoost Performance:

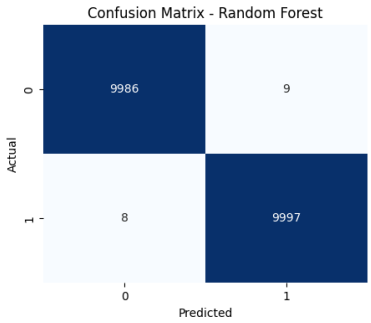
	precision	recall	f1-score	support
0	1.00	1.00	1.00	9995
1	1.00	1.00	1.00	10005
accuracy			1.00	20000
macro avg	1.00	1.00	1.00	20000
weighted avg	1.00	1.00	1.00	20000

AUC-ROC: 0.9999011849752962

Logistic Regression Performance:

	precision	recall	f1-score	support
0	0.95	0.98	0.97	9995
1	0.98	0.95	0.97	10005
accuracy			0.97	20000
macro avg	0.97	0.97	0.97	20000
weighted avg	0.97	0.97	0.97	20000

AUC-ROC: 0.9934971833742958



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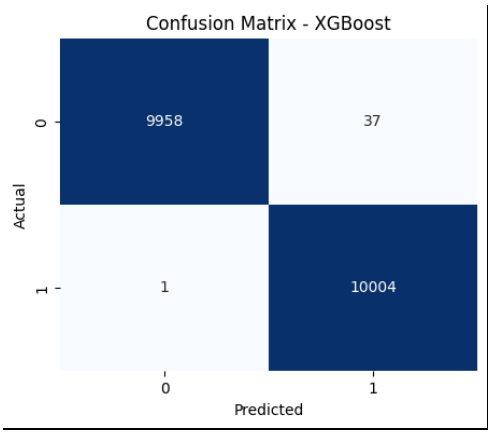

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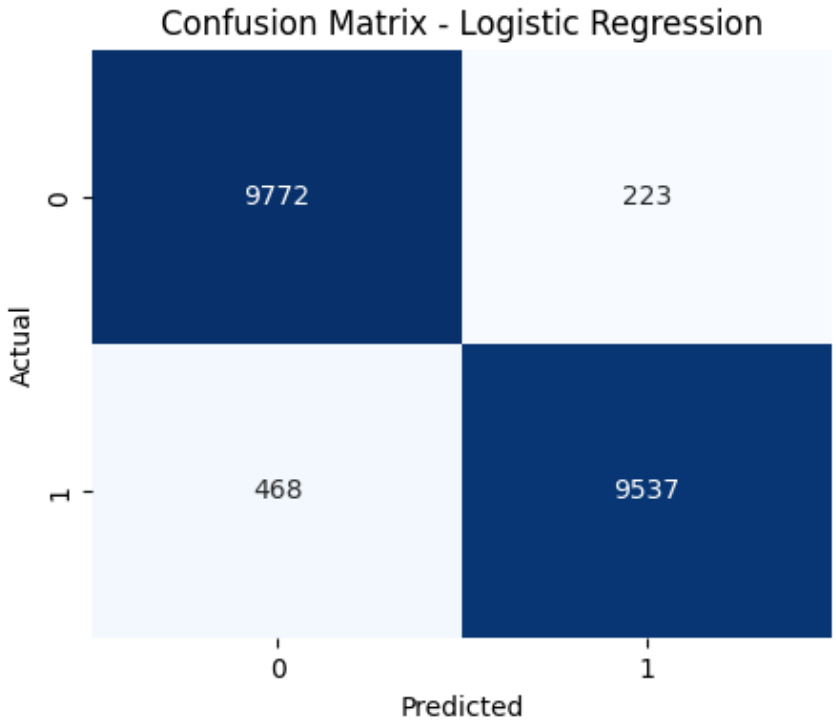
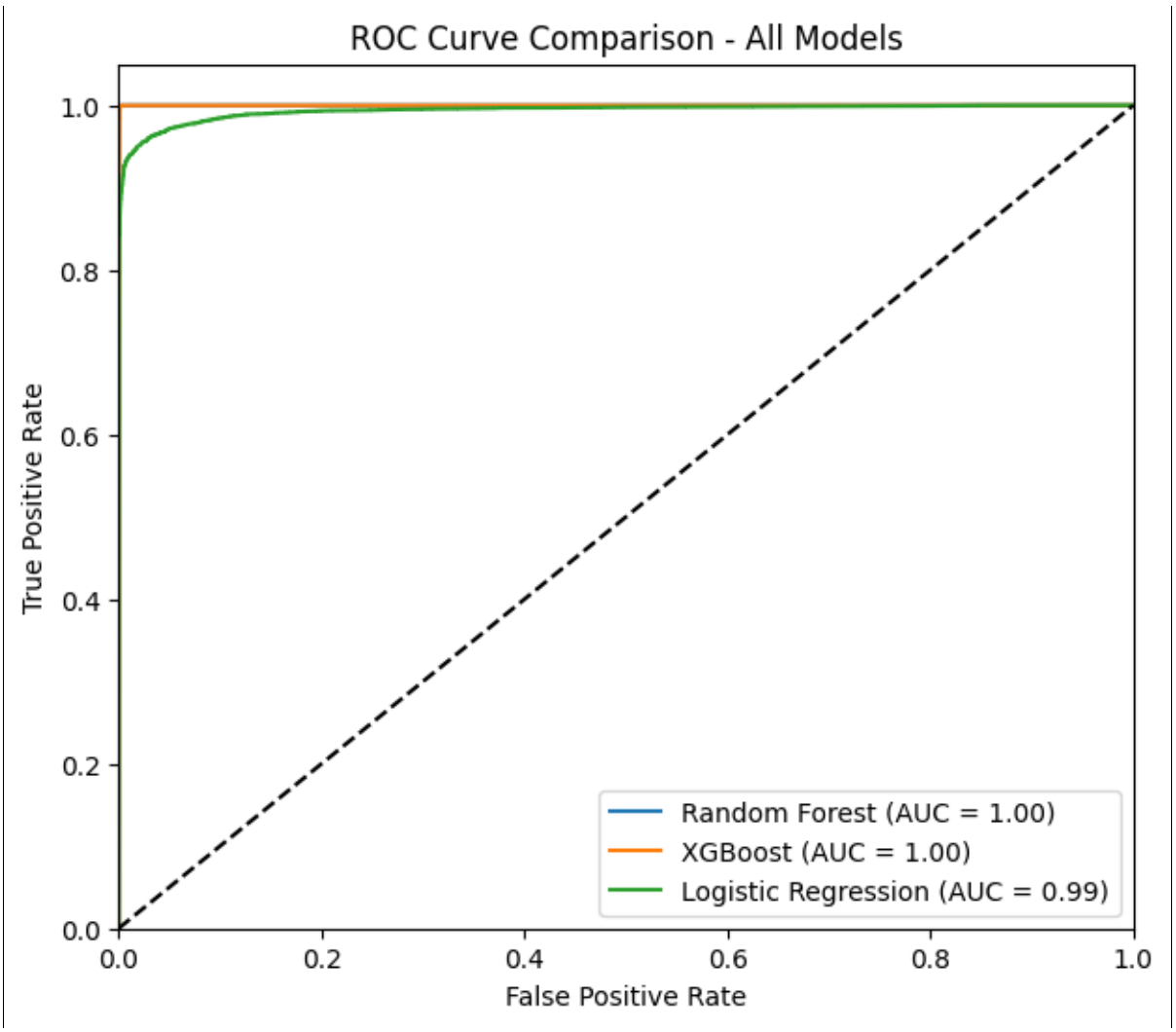
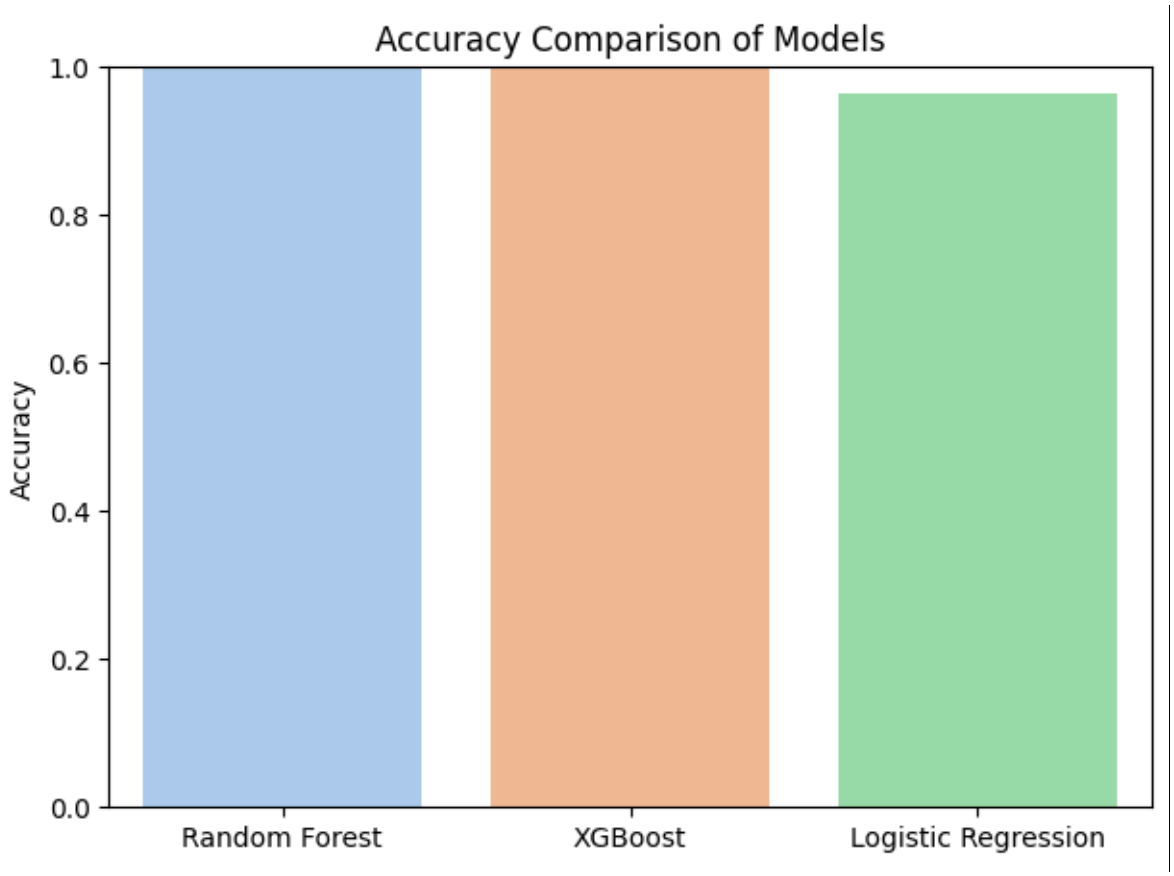

Confusion Matrix - Logistic Regression
9772
223
468
9537
0
1
Actual
0
1
Predicted


ROC Curve Comparison - All Models
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Random Forest (AUC = 1.00)
XGBoost (AUC = 1.00)
Logistic Regression (AUC = 0.99)


Accuracy Comparison of Models
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
Random Forest
XGBoost
Logistic Regression

[redacted]

```
Starting Real-Time Stream Simulation...

Txn 001: Fraud=False, Probability=0.0007, True=0
Txn 002: Fraud=False, Probability=0.0002, True=0
Txn 003: Fraud=True, Probability=0.9998, True=1
Txn 004: Fraud=True, Probability=0.9998, True=1
Txn 005: Fraud=False, Probability=0.0002, True=0
Txn 006: Fraud=True, Probability=1.0000, True=1
Txn 007: Fraud=False, Probability=0.0003, True=0
Txn 008: Fraud=True, Probability=0.9998, True=1
Txn 009: Fraud=True, Probability=0.9999, True=1
Txn 010: Fraud=False, Probability=0.0001, True=0
Txn 011: Fraud=True, Probability=1.0000, True=1
Txn 012: Fraud=True, Probability=0.6098, True=1
Txn 013: Fraud=False, Probability=0.0010, True=0
Txn 014: Fraud=True, Probability=0.9993, True=1
Txn 015: Fraud=False, Probability=0.0000, True=0
Txn 016: Fraud=False, Probability=0.0011, True=0
Txn 017: Fraud=False, Probability=0.0161, True=0
Txn 018: Fraud=True, Probability=0.9994, True=1
Txn 019: Fraud=True, Probability=0.9999, True=1
Txn 020: Fraud=True, Probability=0.9680, True=1
Txn 021: Fraud=False, Probability=0.0001, True=0
Txn 022: Fraud=True, Probability=0.9998, True=1
Txn 023: Fraud=True, Probability=0.9999, True=1
Txn 024: Fraud=False, Probability=0.0086, True=0
Txn 025: Fraud=True, Probability=0.9999, True=1
Txn 026: Fraud=True, Probability=1.0000, True=1
Txn 027: Fraud=False, Probability=0.0002, True=0
Txn 028: Fraud=True, Probability=0.9994, True=1
Txn 029: Fraud=True, Probability=1.0000, True=1
Txn 030: Fraud=True, Probability=0.9999, True=1
Txn 031: Fraud=True, Probability=0.9999, True=1
Txn 032: Fraud=True, Probability=1.0000, True=1
Txn 033: Fraud=True, Probability=1.0000, True=1
Txn 034: Fraud=True, Probability=0.9998, True=1
Txn 035: Fraud=True, Probability=0.9999, True=1
Txn 036: Fraud=False, Probability=0.0007, True=0
Txn 037: Fraud=True, Probability=0.9999, True=1
Txn 038: Fraud=False, Probability=0.0000, True=0
Txn 039: Fraud=False, Probability=0.0001, True=0
Txn 040: Fraud=True, Probability=0.9999, True=1
Txn 041: Fraud=True, Probability=0.9993, True=1
Txn 042: Fraud=False, Probability=0.0001, True=0
```

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